GTÜ
PHYS 121 2019-2020 Fall Semester
First Midterm
90 Minutes

1	2	3	4	5	Total

Name: ..... **Student No:** ..... **/Lecturer**.....

You can use calculator during the exam but exchanging is not allowed.
Take g = 9,80 m/s^2 if necessary. Each question worth 20 points. **Good luck.**

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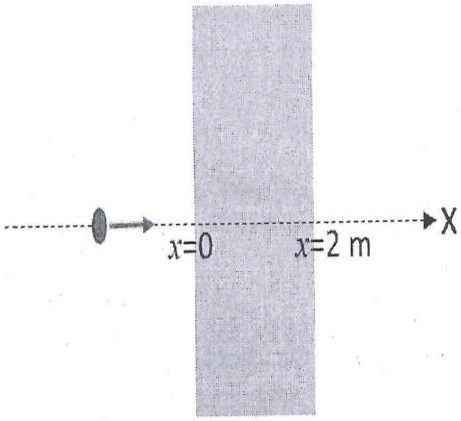
1. A particle moving on a straight line along *x*-axis enters a dense medium at *t*=0 and slows down with a velocity of $v_x(t) = e^{-t}$ m/s. The medium occupies the region between *x*=0.0 m and *x*=2.0 m.

(a) Find the acceleration of this particle a_x, and show that $a_x = -bv_x$, where b is a constant. What is the value and the unit of b?

(b) What is the position of this particle at 2.0 s after it enters the medium?

(c) Can this particle reach the other side of the dense medium at *x*=2.0 m?

(d) Info: ($\int_0^t e^{-t'}dt' = 1 - e^{-t}$), ($\frac{d}{dt}e^{-t} = -e^{-t}$)



a) $a_x = \frac{dv_x}{dt} = \frac{de^{-t}}{dt} = -e^{-t}$

3 puan

$= -bv_x$

1 puan

1 puan

$b = 1$.

birimi $= \frac{m/s^2}{m/s} = \frac{1}{s}$

3 puan

b) $x(t) = x_0 + \int_0^t e^{-t'}dt' = 0 + \int_0^t e^{-t'}dt' = 1 - e^{-t}$

4 puan

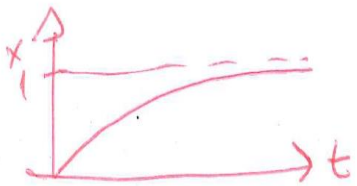
$x(2) = 1 - e^{-2}$ m

$= 0,87$ m

2 puan

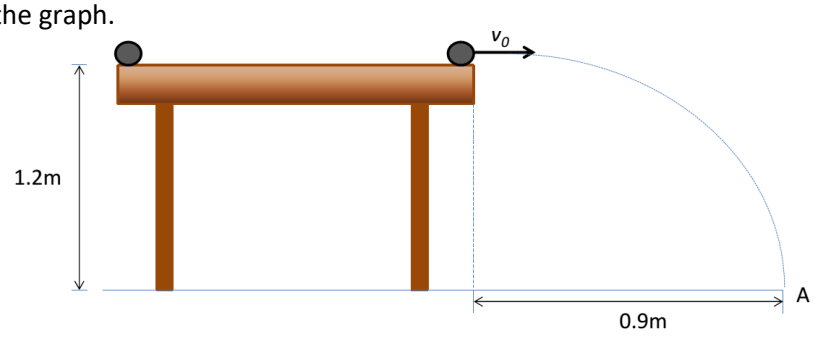
c) $t \to \infty$, $x(t) \to 1$, it can not reach other side

6 puan



2. A ball initially at rest rolls across a table with acceleration rate of 0.5 m/s^2. The ball leaves the table with a speed of v_0 and lands 0.9 m away from the edge of the table.

a. Find the time it takes the ball to reach position A after it leaves the edge of the table as in the figure.
b. Find the length of the table
c. Draw the horizontal velocity (v_x) versus time (t) graph for the entire trip of the ball. Please label the axes of the graph.



a) Height that it falls:

$h = \frac{1}{2} g t^2$ $\quad 1.2 = \frac{1}{2} \cdot 9.8 \times t^2$ $\quad t = 0.49\ s$ 5

b) After leaving the table, distance that it travels on the horizontal axis:

$R = 0.9\ m$ $\quad R = V_0 . t$ $\quad 0.9 = V_0 . 0.49$ $\quad V_0 = 1.81\ m/s$ 5

As it moves on the table with $a = 0.5\ m/s^2$ it reaches $V_0 = 1.81\ m/s$ speed and travels the distance D on the table.

$D = \frac{1}{2} a t^2$

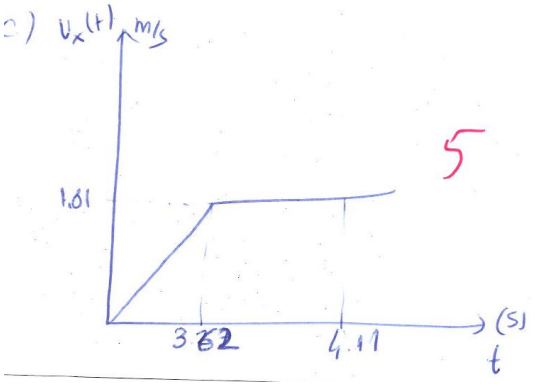
$V_0 = V_{int.} + a t$ $\quad V_0 = a t$ $\quad 1.81 = 0.5 \times t$ 2.5

at rest

$t = 3.62\ sec$

$D = \frac{1}{2} a t^2 = \frac{1}{2} \times 0.5 \times (3.62)^2 = 3.276\ m$ 2.5

c) $V_x(t)$ m/s

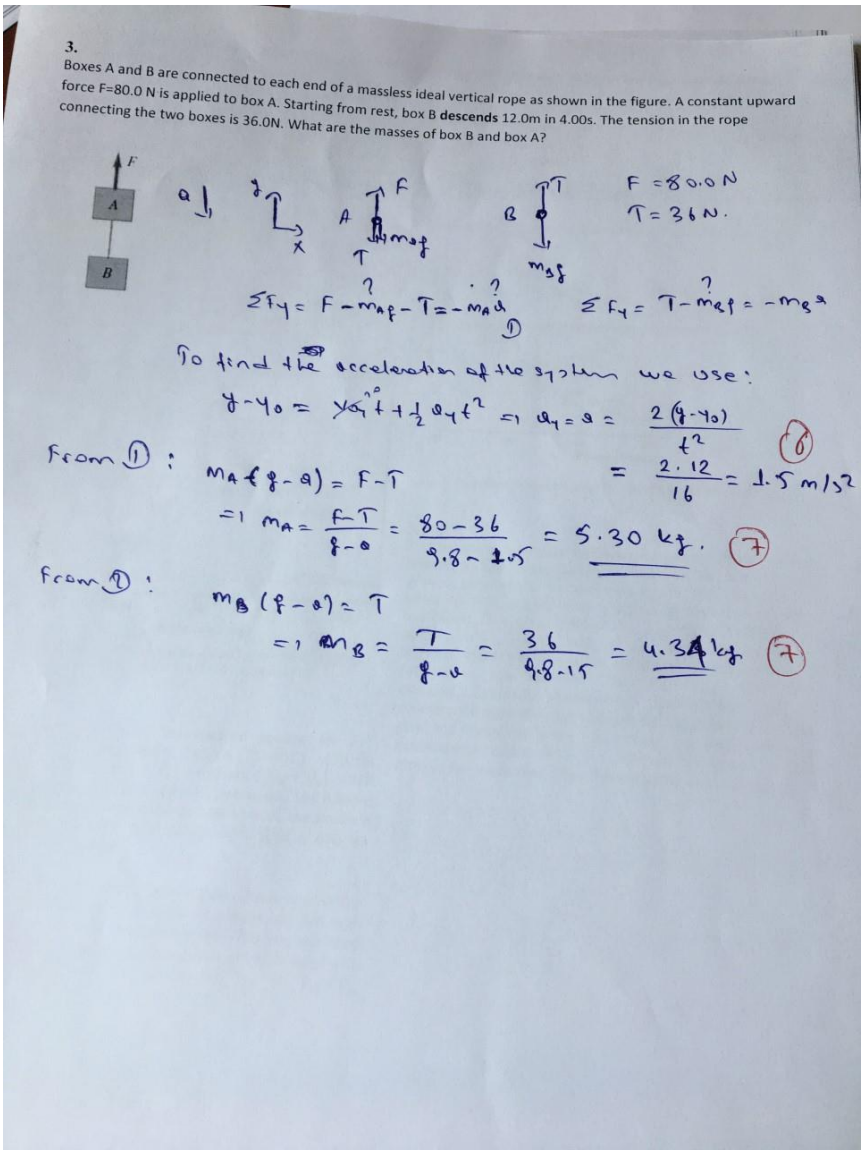


5

3.

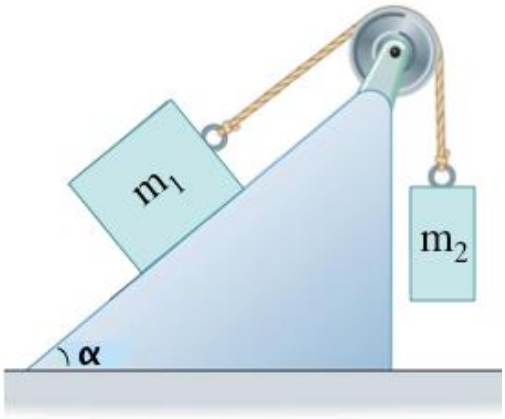
Boxes A and B are connected to each end of a massless ideal vertical rope as shown in the figure. A constant upward force F=80.0 N is applied to box A. Starting from rest, box B **descends** 12.0m in 4.00s. The tension in the rope connecting the two boxes is 36.0N. What are the masses of box B and box A?



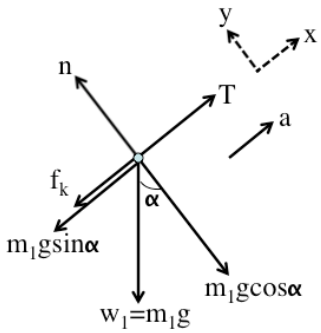


4. A block with mass m_1 = 20.0 kg is placed on an inclined plane with slope angle α = 53.1° and is connected to a hanging block with mass m_2 = 36.0 kg by a cord passing over a small, frictionless pulley. The block with mass m_1 **moves up** the plane. The coefficient of static friction between the surfaces is $\mu_s = 0.50$ and the coefficient of kinetic friction between the surfaces is $\mu_k = 0.40$

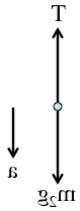
a) Draw free body diagrams for both blocks.
b) Find the magnitude of the acceleration of the blocks.
c) Find the tension in the string.



(4 puan)



(2 puan)



b) $\sum F_y = n - m_1 g cos\alpha = 0$ (1 puan)

$n = m_1 g cos\alpha$

$f_k = \mu_k n = \mu_k m_1 g cos\alpha$ (2 puan)

$\sum F_x = T - f_k - m_1 g sin\alpha = m_1 a$ (3 puan)

Let's put T and f_k into this equation

$m_2 g - m_2 a - \mu_k m_1 g cos\alpha - m_1 g sin\alpha = m_1 a$

$m_2 g - \mu_k m_1 g cos\alpha - m_1 g sin\alpha = m_1 a + m_2 a$

$a = \frac{m_2 g - \mu_k m_1 g cos\alpha - m_1 g sin\alpha}{m_1 + m_2}$

$= \frac{(36.0kg)(9.80m/s^2) - (0.40)(20.0kg)(9.80m/s^2)(cos53.1°) - (20.0kg)(9.80m/s^2)(sin53.1°)}{20.0kg + 36.0kg}$

$a = 2.66\, m/s^2$ (3 puan)

$\sum F_y = m_2 g - T = m_2 a$

$T = m_2 g - m_2 a$

c) From part a,

$\sum F_y = m_2 g - T = m_2 a$ (2 puan)

$T = m_2 g - m_2 a$

so put the value of a in this equation,

$T = m_2(g - a)$

$$T = 36.0kg(9.80\, m/s^2 - 2.66\, m/s^2)$$

$T = 257.1\ Newton$ (3 puan)

5. A constant force $\vec{F} = (-68.0\ N)\hat{\imath} + (36.0N)\hat{\jmath}$ is applied to a 380-kg car as the car travels 48.0 m in a direction that is 240.00 counterclockwise from the +x-axis.

a) What is the acceleration of the car?

(b)How much work does the applied force do on the car?

$\vec{F} = m\vec{a}$

$\vec{a} = \vec{F}/m = \frac{(-68.0\ N)\,\hat{\imath} + (36.0N)\hat{\jmath}}{380\ kg} = (-0.179\ m/s^2)\hat{\imath} + (0.0947\ m/s^2)\hat{\jmath}$ (6p)

(b)

$W = \vec{F}.\vec{s} = F_x s_x + F_y s_y$ (3p)

Components of the $\vec{s}$ are *x=(48.0m).cos240.* 0^0=-24.0 m (3p)

y=(48.0m).sin240. 0^0=-41.57 m

Thus;

$\vec{s} = (-24.0\ m)\hat{\imath} + (-41.57\ m)\hat{\jmath}$

The definition of work gives:

$W = \vec{F}.\vec{s} = (-68.0\ N)(-24.0\ m)\hat{\imath}.\hat{\imath} + (36.0N)(-41.57\ m)\hat{\jmath}.\hat{\jmath} = (+1632\ J) + (-1497\ J) = +135\ J$ (8 p)